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Correct Answer

Next Q

Practice Exam - 1

Question 90 of 117



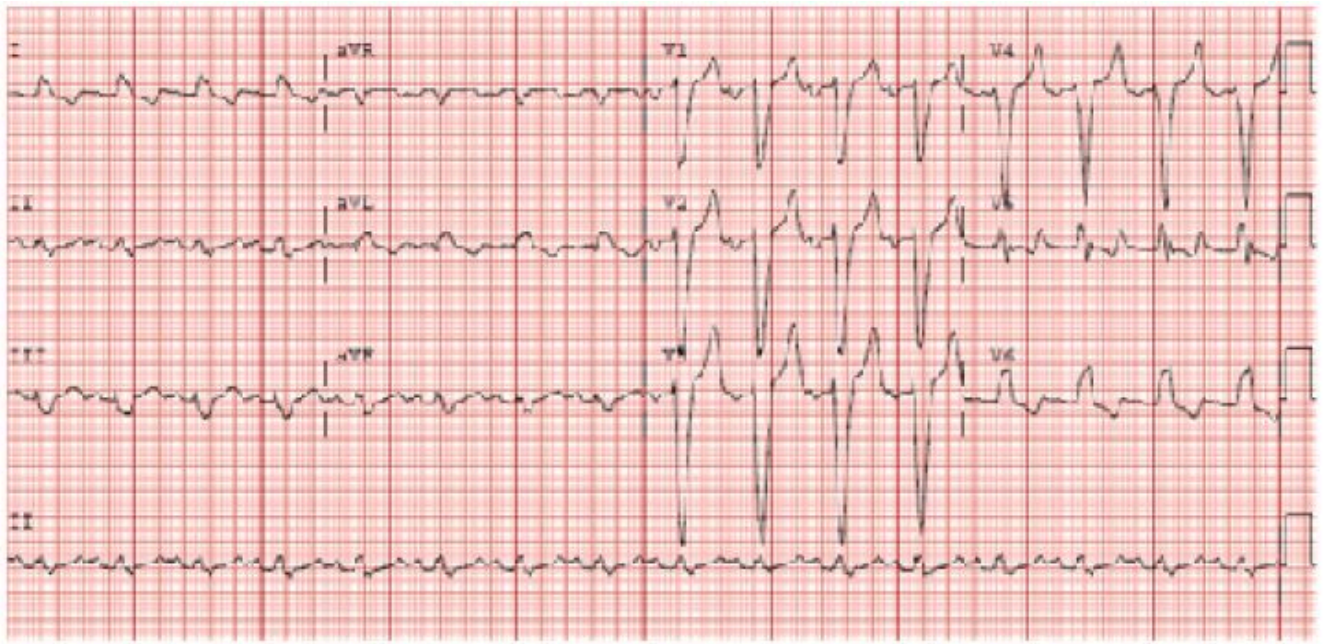
A 68-year-old female with a nonischemic cardiomyopathy (NICM), mild chronic kidney disease (CKD), hypertension, diabetes and obesity is referred to you for evaluation for ventricular assist device (VAD) candidacy.

The patient reports a longstanding history of mild dyspnea on exertion (DOE) for the preceding 10 years. In the past year, she noted some worsening of her DOE in addition to mild orthopnea and fatigue. She was seen by her primary care physician and eventually underwent an echocardiogram (TTE) that revealed a left ventricular end-diastolic dimension (LVEDD) of 6.3 cm, an ejection fraction (EF) of 20% and moderate to severe mitral regurgitation (MR). Coronary angiogram showed no significant coronary artery disease. She was referred to a cardiologist and was initiated on appropriate guideline directed medical therapy (GDMT) with carvedilol 25 mg twice daily, sacubitril-valsartan 97-103 mg twice daily, dapagliflozin 10 mg and spironolactone 25 mg daily.

She was also initiated on maintenance diuretic with bumetanide 1 mg daily. With this regimen, her symptoms improved but did not resolve. Repeat TTE after 6 months of GDMT demonstrated an EF of 30% with moderate MR and she was referred to your clinic.

In clinic today, her blood pressure is 103/74 mmHg, heart rate is 68 bpm, oxygen saturation is 98% on room air. Her physical exam is only notable for a JVP of 7 cm of H₂O with positive HJR, regular rate and rhythm with III/VI holosystolic murmur radiating to the apex. The remainder of her physical exam is unrevealing. Her blood work is notable for a sodium of 136 mmol/L, potassium of 4.7 mmol/L, creatinine of 1.5 mg/dL. CBC, LFTs and TSH are unremarkable.

The EKG shows: Which is the best next step?



EKG

- A.** Initiate evaluation for ventricular assist device
- B.** Initiate evaluation for heart transplantation
- C.** Referral to electrophysiology for cardiac resynchronization therapy (CRT) with a defibrillator
- D.** Referral to interventional cardiology for MitraClip

References

1. Yancy et al. 2013 ACCF/AHA Guideline for the management of Heart Failure. Circulation 2013; 128; e240-e327
2. Stone et al. Transcatheter Mitral-Valve Repair in Patients with Heart Failure. NEJM 2018; 379 (24): 2307- 2318
3. Zusterzeel et al. Cardiac resynchronization therapy in women: US Food and Drug Administration meta- analysis of patient-level data. JAMA Int Medicine. 2014 Aug;174(8):1340-8.
4. van Bommel et al. Characteristics of heart failure patients associated with good and poor response to cardiac resynchronization therapy: a PROSPECT (Predictors of Response to CRT) sub-analysis. Eur Heart J. 2009 Oct;30(20):2470-7